

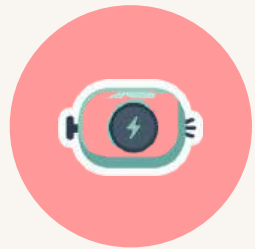
THE MAGIC SWITCH!

MAKING MOTORS DANCE!



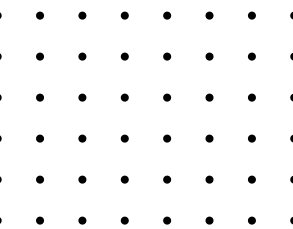
THE MAGIC SWITCH

WHAT IS A MOTOR?



MAGICAL SPINNERS THAT MAKE THINGS MOVE!

Motors are amazing devices that turn electricity into spinning motion! They're like magical spinners hiding inside many things you use every day. Your toy cars zoom because of motors. Ceiling fans spin to keep you cool because of motors. Even the washing machine that cleans your clothes has a motor inside!



THE BIG QUESTION!



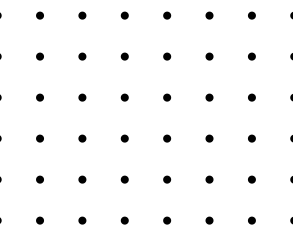
How can we make a motor spin forward?
Think about your toy cars - what makes the wheels go one way?



How can we make a motor spin backward?
Can the same motor go both directions?
Let's find out!



THINK!

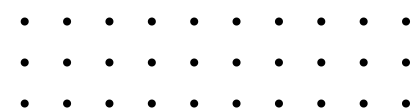
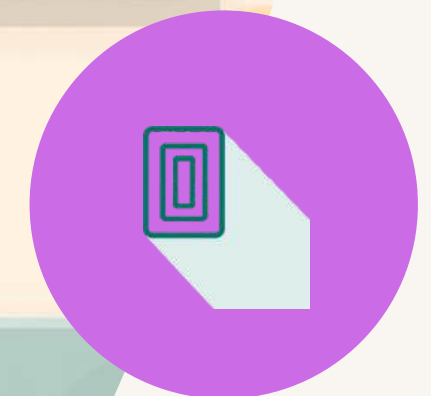
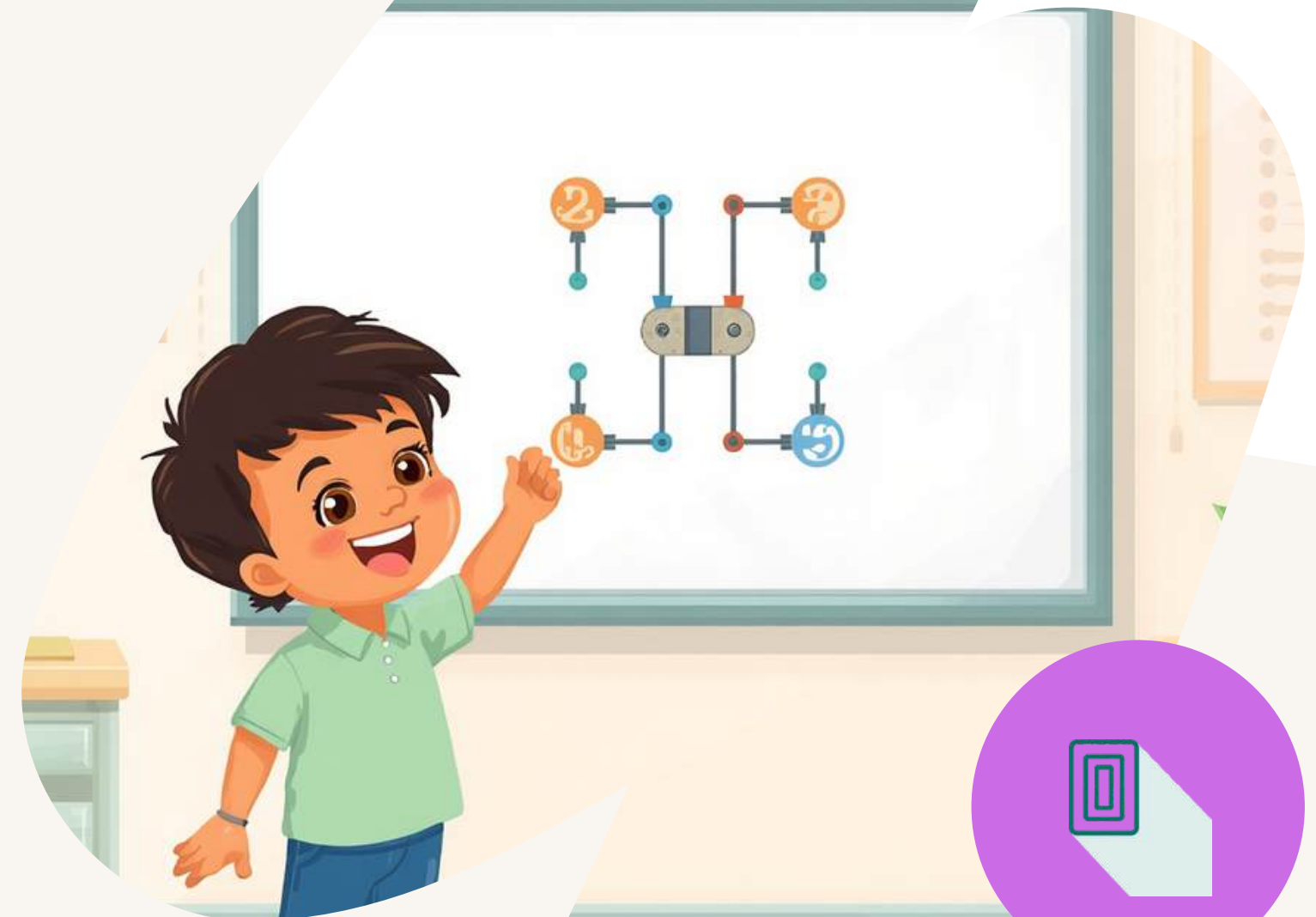


THE H-BRIDGE

YOUR MAGIC SWITCH!

The H-Bridge is a special circuit that looks like the letter 'H'! It has 4 switches that work together like a team.

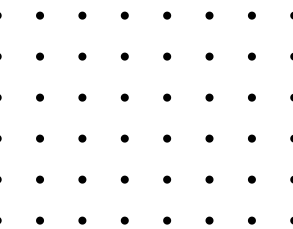
These 4 switches control which way electricity flows to your motor. By flipping different switches, you can make the motor spin forward or backward!



GOING FORWARD!

> HOW IT WORKS

When we flip the switches in one special pattern, electricity flows in one direction through our H-Bridge circuit. This makes the motor spin forward - just like when a car moves ahead on the road! The current travels from positive to negative in a straight path.



Build Your Magic Switch!

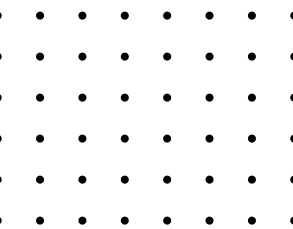
MATERIALS NEEDED:

- 1 DC Motor
- Battery
- Switches
- Connecting Wires (Red for Positive, Black for Negative)
- Cardboard base



GOING BACKWARD!

Here's the cool part - flip the switches differently, and the electricity flows the opposite way through the H-Bridge! When the current goes in reverse, the motor spins backward. It's like magic - same motor, same battery, but now it moves in the other direction!



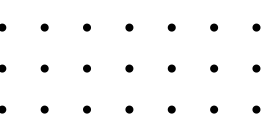
STEPS TO MAKE MOTORS DANCE:

Step 1: Tape your motor to the cardboard base. Attach a small piece of tape to the motor shaft

Step 2: Connect your battery to the switches. Remember: Red is Positive (+) and Black is Negative (-).

Step 3: Creating the “H” path by wiring the switches so they form a path for the electricity. One pattern of switches must connect the battery to the motor in one way, and the other pattern must flip the connections.

Step 4: Test Your Magic! Flip the switches back and forth and see which way the motor spins.



WHAT DID WE LEARN?



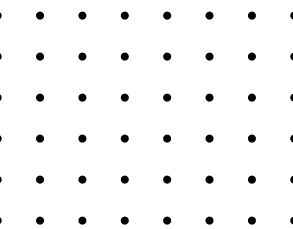
H-Bridge has 4 special switches that work together to control which way your motor spins - like a magic control panel!



Different switch patterns make different things happen - some patterns make the motor go forward, others make it go backward!



100%
COMPLETE!



THE MAGIC SWITCH

GREAT JOB, YOUNG ENGINEERS!

YOU LEARNED HOW TO MAKE MOTORS DANCE
FORWARD AND BACKWARD!

